Chinmay Mahananda

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Education

NORTHEASTERN UNIVERSITY — Seattle, WA
Master of Science in Electrical and Computer Engineering, GPA: 3.92/4.0 — December 2027
Concentration: Hardware and Software for Machine Intelligence
Relevant Coursework: VLSI Design, Computer Architecture (in progress), Fundamentals of Computer Engineering (in progress), Machine Learning and Pattern Recognition, Computer Vision, Advanced Computer Vision
Seattle Campus Scholarship and International Impact Award (merit-based, 30% of program tuition).
Available for co-op starting January 2027 (4, 6, or 8 months).

DR. AMBEDKAR INSTITUTE OF TECHNOLOGY — Bengaluru, India
Bachelor of Engineering in Electronics and Communication Engineering — October 2024

Technical Skills

Languages: Verilog, SystemVerilog, Python, Embedded C, MATLAB, SPICE, C++
EDA Tools: Yosys, OpenROAD, OpenLane, Sky130 PDK, Icarus Verilog, Cadence Virtuoso, Spectre, LTspice, KiCad, Altium
Design & Verification: RTL design, logic synthesis, place and route, static timing analysis, constrained-random and self-checking testbenches, functional coverage, golden-model co-simulation, waveform debug, Git, GitHub Actions CI, Linux
Currently Learning: UVM 1.2, SystemVerilog Assertions (SVA), Vivado xsim

Projects

RISC-V Pipelined Core: RTL-to-GDSII on Sky130 — July 2026
Signed off a 4-stage pipelined RV32I-subset core to GDSII at 333 MHz on Sky130 using Yosys and OpenROAD (WNS/TNS 0.0 ns, zero DRC/LVS/routing violations, 613,701 µm² core), and documented a 500 MHz attempt that failed timing. Designed the Verilog RTL (IF, EX, MEM, WB) with forwarding, load-use stalling, and branch squashing, verified bit-exact against a Python golden model and assembler written from scratch. github.com/chinmaymahananda/riscv-pipelined-core

INT8 CNN Inference Accelerator: Systolic MAC Array — June 2026 – September 2026
Achieved 20/20 bit-exact end-to-end inference against a Python INT8 golden model (18/20 classified correctly vs. 89.7% FP32 test accuracy) on a weight-stationary three-layer CNN accelerator (Conv1, Conv2, FC), built in Verilog from a parameterizable output-stationary systolic MAC array with shift-only power-of-two requantization. Verified with a SystemVerilog constrained-random testbench whose functional coverage fails on any uncovered bin (206 cases, 0 mismatches, 0 coverage holes) and a 1,900-case Python model sweep from 3×3 to 16×16. github.com/chinmaymahananda/systolic-mac-array

mxgold: Golden Reference Model for OCP Microscaling (MX) Arithmetic — August 2026 – October 2026
Built a bit-exact Python golden reference for MX arithmetic passing 43/43 conformance checks (v0.1.0, CI on Python 3.10–3.13): zero dependencies, policy-parameterized, returning raw bit patterns for block quantization and dot products across MXFP8 (E4M3/E5M2), MXFP4, and NVFP4. Reproduced two documented vendor divergences (E4M3 overflow policy; MXFP4 vs. NVFP4 block structure) and documented a third the spec leaves open, block-scale selection, where two conformant encoders differ by 0.1892 vs. 0.0000 relative error on the same block. github.com/chinmaymahananda/mx-conformance

Experience

APPSNDEVICES TECHNOLOGIES PRIVATE LIMITED — Bengaluru, India (Remote)
STEM Content Developer (Part-time) — May 2025 – August 2025
Built interactive physics and electronics simulations (field behavior, circuit response, wave propagation) packaged into teaching modules deployed through state government education programs in Assam and Uttarakhand, reaching 10,000+ students.

SRI CHAITANYA TECHNO SCHOOL — Bengaluru, India
Physics Faculty, IIT-JEE and NEET Preparation (Part-time) — September 2024 – November 2024
Taught exam-preparation physics (electromagnetism, circuits); received the Best Teacher Award (2024).

PRIVATE TUTORING — Bengaluru, India
Physics, Mathematics, and Chemistry Tutor (Part-time) — August 2020 – August 2024
Tutored 300+ ICSE and CBSE students over four years; batch exam scores improved 25%+ on average.

Publications & Certifications

"Design of 6G Antenna," International Journal of Novel Research and Development (IJNRD), Vol. 9(6), June 2024. First author.
Emertxe Certified Embedded Professional (ECEP), Emertxe Information Technologies, August 2025. Credential ID 24032F_023.